Predicting Earthquake Damage to Prioritise Building Inspections

Logistic regression, random forest and gradient boosting on the 2015 Gorkha earthquake building-damage data (Richter's Predictor)

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Abstract

After the 2015 Gorkha earthquake, Nepal surveyed hundreds of thousands of damaged buildings. We ask whether a building's recorded structure and location can predict its damage grade well enough to rank buildings for inspection. Using 260,601 labelled buildings from the DrivenData competition Richter's Predictor, we built a leakage-safe scikit-learn pipeline and compared three classifiers: logistic regression, a random forest and HistGradientBoosting. Seven controlled logistic-regression experiments showed that location, encoded with cross-fitted multiclass target encoding of 11,595 local areas, gave the largest single gain (+6.3 pp in cross-validated micro-F1). An ordinal formulation did not outperform the multinomial model, and class weighting traded overall accuracy for recall of the rare low-damage class. The tuned logistic model reaches a holdout micro-F1 of 0.740 (macro-F1 0.680), against 0.569 for the majority-class baseline. With the same features and folds, the tuned random forest reaches 0.750 and HistGradientBoosting 0.750. HistGradientBoosting, which had the highest cross-validated micro-F1, is our final model: it gains +0.98 pp in cross-validation and +0.93 pp on the holdout over logistic regression. Ranking buildings by the final model's probability of grade-3 damage, inspecting the top 20% finds 50.6% of all grade-3 buildings, against 20% for a random order. The logistic coefficients agree with engineering intuition: mud-mortar stone and stone-flag construction raise the odds of destruction, and engineered reinforced concrete and cement-bonded masonry lower them.

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1. Introduction and Problem Statement

On 25 April 2015 a magnitude-7.8 earthquake struck the Gorkha district of Nepal. It killed nearly 9,000 people and damaged or destroyed hundreds of thousands of buildings (National Planning Commission, 2015). After a disaster on this scale, trained structural inspectors are a scarce resource. Every building must eventually be assessed, but the order of inspection decides how quickly dangerous structures are cordoned off, families are rehoused, and reconstruction aid is targeted.

The Government of Nepal later released one of the largest post-disaster building surveys ever collected. A version of it underlies the DrivenData competition *Richter's Predictor: Modeling Earthquake Damage* (DrivenData, 2019). This project uses those data to answer one question:

> *"Can we predict how badly a building was damaged in the 2015 Gorkha earthquake from its structure and location, so post-disaster inspectors can prioritise which buildings to assess first?"*

We treat this as a three-class classification problem with an ordered target: damage grade 1 (low), 2 (medium) and 3 (almost complete destruction). We compare three classification models. Model 1 is **logistic regression**, which we study in most depth: its predictions are fast, its coefficients can be audited by engineers, and its probabilities can be used directly to rank buildings. Models 2 and 3 are tree ensembles, a **random forest** and **HistGradientBoosting**, which can learn interactions and non-linear effects that a linear model cannot. Our contributions are:

- a fully leakage-safe preprocessing and modelling pipeline, evaluated with stratified 5-fold cross-validation and a holdout that each tuned model sees only once;
- a controlled sequence of seven logistic-regression experiments that isolates the value of structure, coarse location, fine location, engineered features, class weighting and an ordinal formulation;
- cross-fitted multiclass target encoding of the high-cardinality location identifiers, which turns out to be the single most valuable idea for all three models;
- a like-for-like comparison of the three tuned models on the same features and folds, with the final model chosen by cross-validation;
- a prioritisation analysis that converts model output into an inspection policy, and an interpretation that combines odds ratios with permutation importance.

2. Dataset Description

The labelled data consist of two files joined on *building_id*: *train_values.csv* with 260,601 rows and 38 features, and *train_labels.csv* with the damage grade. A further 86,868 buildings in *test_values.csv* have no labels; we use them only to produce a competition submission. Table 1 groups the features.

Table 1. Feature groups in the Richter's Predictor data.

Group	Features	Type
Location	geo_level_1_id (31 regions), geo_level_2_id (1,414), geo_level_3_id (11,595)	nested categorical ids
Size & age	age, area_percentage, height_percentage, count_floors_pre_eq	numeric (normalised)
Construction	foundation_type, roof_type, ground_floor_type, other_floor_type, position, plan_configuration, land_surface_condition	categorical (anonymised letters)
Superstructure	11 has_superstructure_* flags (adobe mud, mud-mortar stone, stone flag, cement-mortar stone/brick, mud-mortar brick, timber, bamboo, RC non-engineered, RC engineered, other)	binary
Use & ownership	legal_ownership_status, count_families, 11 has_secondary_use_* flags	categorical / numeric / binary

We verified the facts given in the brief rather than assuming them, and the notebook asserts each one. There are 0 missing values in either file, 8 categorical columns and 22 binary columns. The class distribution is imbalanced: grade 1 9.6%, grade 2 56.9% and grade 3 33.5% (Figure 1). Two data-quality issues matter. First, 1,390 buildings (0.53%) have *age = 995*, a placeholder far beyond the largest real age of 200 years. Second, the finest location level has 11,595 distinct values, too many to one-hot encode sensibly. Also, 0.38% of test buildings fall in a geo_level_3 area that never appears in the labelled data.

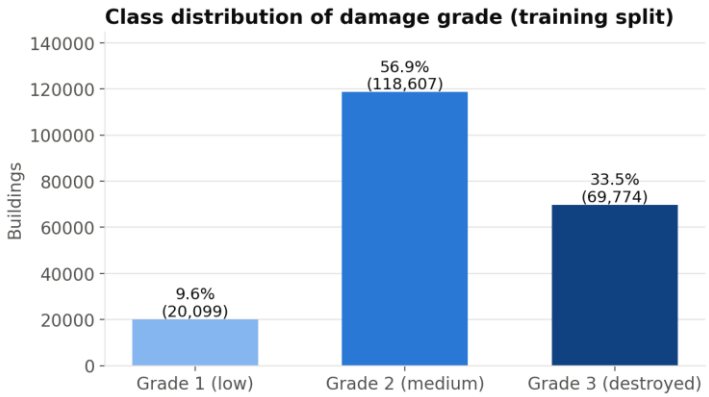


Figure 1. Damage-grade distribution in the 80% training split. Grade 2 is the majority class; grade 1 is rare.

3. Data Pre-processing

Before any exploration, we made a stratified 80/20 split (208,480 training and 52,121 holdout buildings). All exploration below uses only the training split, so nothing we decided about the features was influenced by the holdout.

3.1 Exploration

Construction type separates the classes sharply (Appendix C, Figure C1). Foundation type r, used by 84% of buildings, has a grade-3 share of 38%. Foundation i has a grade-3 share of only 2% and a grade-1 share of 57%. Roof type x shows the same protective pattern (4% grade 3). The category codes are anonymised, but these profiles are consistent with engineered, reinforced construction.

Superstructure material is the clearest structural signal (Figure 2). Mud-mortar stone is used in 76% of buildings, and 38% of those reached grade 3. Engineered reinforced concrete (RC) reached grade 3 in only 1.9% of cases, with 64% at grade 1. Cement-mortar brick is similarly protective (6% grade 3).

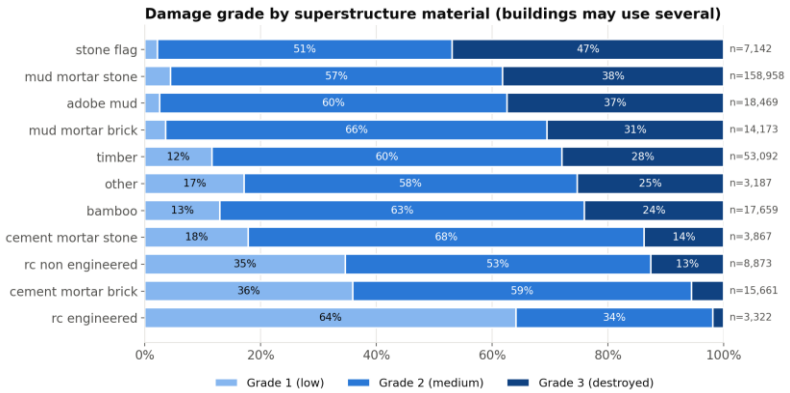


Figure 2. Damage-grade mix by superstructure material. A building can use several materials, so rows overlap.

Location matters at least as much as structure (Figure 3). Across the 31 top-level regions, the grade-3 share ranges from 9% (region 26) to 81% (region 17). Location most likely stands in for shaking intensity, that is, distance from the epicentre and local ground conditions. These are not in the data, but they strongly shape damage.

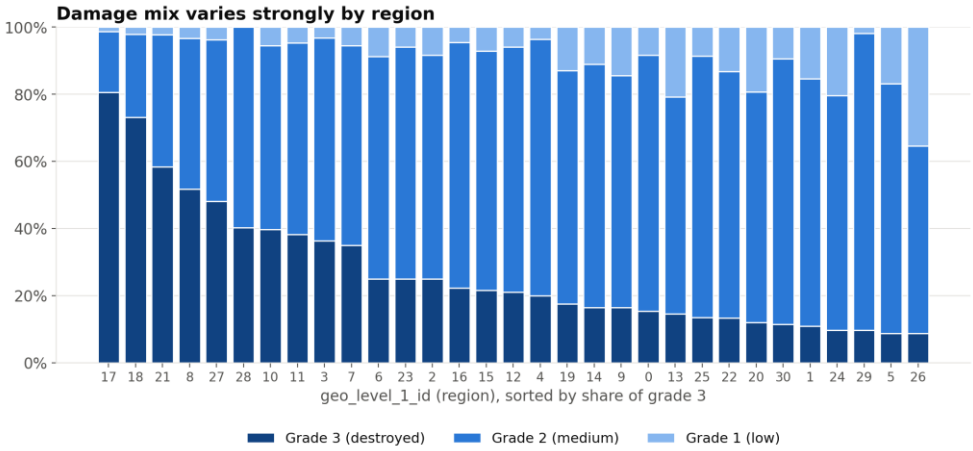


Figure 3. Damage-grade mix for each of the 31 geo_level_1 regions, sorted by grade-3 share.

The numeric features are right-skewed, with skewness 2.05 for age, 3.52 for area and 1.73 for height. A log1p transform reduces this to -1.04, 0.41 and -0.15 (Appendix C, Figure C2). Individually these features are only weakly related to damage: the largest absolute Spearman correlation with damage grade among numeric and material features is 0.27. Height and number of floors are strongly collinear ($\rho = 0.76$; Appendix C, Figure C3). The weak marginal correlations show that damage comes from combinations of many weak signals, which is what a multivariate model needs to capture. The L2 penalty of logistic regression keeps the collinear pair stable.

3.2 Cleaning

No imputation of genuinely missing values was needed. The age placeholder was handled explicitly. Buildings with age = 995 have a lower grade-3 share (28.6%) than the rest (33.5%), so the placeholder carries information and should not be treated as a real age of 995 years. A custom transformer (AgeCleaner) therefore (i) adds a binary indicator age_is_placeholder and (ii) replaces the placeholder with the median age of the training fold. Because the median is learned in fit, it is recomputed inside every cross-validation fold.

3.3 Encoding and scaling

All transformations are implemented as one scikit-learn ColumnTransformer inside a Pipeline (Table 2), so every learned statistic comes from training data only.

Table 2. Preprocessing pipeline. Every step is fitted on training folds only.

Columns	Transformation	Rationale
age, area_percentage, height_percentage	log1p, then StandardScaler	reduce right skew; common scale for L2 and lbfgs
count_floors_pre_eq, count_families (+ engineered)	StandardScaler	small counts, little skew
8 categorical columns	OneHotEncoder (all levels, handle_unknown='ignore')	nominal codes; L2 keeps full dummy sets identifiable
geo_level_1_id (E2+)	OneHotEncoder	31 regions, small enough to one-hot
geo_level_2_id, geo_level_3_id (E3+)	TargetEncoder(target_type='multiclass'), 5-fold cross-fitting	high cardinality; 3 smoothed class rates per level
22 binary flags + age_is_placeholder	passthrough	already 0/1

The multiclass TargetEncoder replaces each geo id by three numbers: the smoothed proportions of grade-1, grade-2 and grade-3 buildings observed for that id. Rare ids are shrunk toward the global class distribution, and ids never seen in training receive the global distribution. Target encoding can leak the label, because a building's own grade would contribute to its own feature. scikit-learn's implementation therefore uses *cross-fitting* in fit_transform: training rows are encoded with statistics computed on the other internal folds (Micci-Barreca, 2001; Pedregosa et al., 2011).

The two tree ensembles receive the same information through a tree-appropriate variant of this pipeline. AgeCleaner, the engineered features and the multiclass TargetEncoder for geo levels 2 and 3 are unchanged. The log transform and scaling are dropped, because tree splits do not change under monotone transformations. The eight categorical columns and geo_level_1_id are integer-coded with an OrdinalEncoder instead of one-hot encoded; unseen codes become missing values, which both ensembles handle natively. HistGradientBoosting treats these nine integer-coded columns as true categories (native categorical splits), while the random forest splits them as ordered codes.

3.4 Feature engineering

Four candidate features were constructed: n_superstructure (number of materials used), log_area_height_ratio (footprint relative to height, a slenderness proxy), floors_x_log_age (taller and older buildings), and height_per_floor. Fold-to-fold noise is of the same order as the expected effects, so each candidate was added to E3 on its own and compared on the same five folds. A candidate was kept only if its mean paired gain was positive *and* it improved at least four of the five folds. Table 3 shows the outcome; the features kept were floors_x_log_age.

Table 3. Add-one ablation of engineered features on top of E3 (E3 micro-F1 = 0.7350).

Candidate	CV micro-F1	Mean paired gain	Folds improved	Decision
n_superstructure	0.7349	-0.00 pp	2/5	dropped
log_area_height_ratio	0.7350	+0.00 pp	2/5	dropped
floors_x_log_age	0.7352	+0.02 pp	5/5	kept
height_per_floor	0.7350	+0.00 pp	1/5	dropped

4. Methodology

4.1 Training and testing procedure

- **Split.** values and labels were merged on building_id and split 80/20, stratified by damage grade, with random_state = 42.
- **Model selection.** every experiment was scored by stratified 5-fold cross-validation (shuffled, random_state = 42) on the 80% split. We report the mean and standard deviation over folds. Identical folds across experiments allow paired comparisons.
- **Tuning.** for logistic regression, the best of E4–E6 by mean CV micro-F1 was tuned over C ∈ {0.01, 0.03, 0.1, 0.3, 1, 3, 10, 30, 100}. The random forest (4 candidates) and HistGradientBoosting (6 candidates) were tuned on small grids (Appendix B). All three used GridSearchCV with the same folds, scoring f1_micro, and then refitted the best setting on the whole 80% split.
- **Final test.** each of the three refitted models was scored once on the 20% holdout; the notebook asserts that no model is scored twice.
- **Final model.** the overall final model is the one with the highest mean CV micro-F1, so the holdout plays no part in the choice.
- **Submission.** the final model (HistGradientBoosting) was refitted on all 260,601 labelled buildings to predict the 86,868 test buildings.

4.2 Models

The multinomial logistic regression (Hosmer et al., 2013; Hastie et al., 2009) models the class probabilities with a softmax over linear scores:

$$P(y = k \mid x) = \exp(\beta_{0k} + \beta_k{}^T x) \,/\, \Sigma_j \exp(\beta_{0j} + \beta_j{}^T x), \quad k \in \{1, 2, 3\}$$

It is fitted by minimising the cross-entropy plus an L2 penalty $\|\beta\|^2/(2C)$ with the lbfgs solver. All inputs are scaled or binary, and we allowed up to 5000 iterations. The largest number of iterations used by any fit (experiments, ablation, all 9×5 grid fits and the refits) was 337. Every fit therefore converged, and no ConvergenceWarning was raised.

For the ordinal model (E6) we follow Frank and Hall (2001). We fit two binary logistic models on the same features, for P(y > 1 | x) and P(y > 2 | x). Class probabilities are differences of the cumulative probabilities:

$$P(y=1) = 1 - P(y>1), \quad P(y=2) = P(y>1) - P(y>2), \quad P(y=3) = P(y>2)$$

The two models are fitted independently, so their cumulative probabilities can occasionally cross. We enforce P(y>2) ≤ P(y>1) before taking differences, and we predict the class with the highest probability. Unlike the proportional-odds model (McCullagh, 1980), the two models do not share slopes.

Model 2, random forest. A random forest (Breiman, 2001) averages the class probabilities of 150 decision trees. Each tree is grown on a bootstrap sample (here of 50% of the training rows, which also bounds memory) and considers only a random subset of features at each split, so the trees are decorrelated and their average has low variance. We tuned the minimum leaf size, which controls tree depth, and the number of features tried per split.

Model 3, HistGradientBoosting. Gradient boosting (Friedman, 2001) builds an additive model of small trees, F_m(x) = F_{m−1}(x) + η·h_m(x), where each new tree h_m is fitted to the gradient of the multiclass log-loss of the current model and η is the learning rate. scikit-learn's HistGradientBoostingClassifier bins every feature into at most 255 bins, which makes 300 boosting iterations on 200,000 rows fast, and splits categorical features

natively. We fixed 300 iterations without early stopping and tuned the learning rate and the maximum number of leaves per tree.

4.3 Experimental design

Each experiment changes one thing relative to its predecessor (Table 4).

Table 4. Experiment design. All logistic models use C = 1 before tuning.

ID	Configuration	Question answered
E0	Majority-class baseline (always grade 2)	What does guessing achieve?
E1	LR, all non-location features	How much does the building itself tell us?
E2	E1 + one-hot geo_level_1_id	Value of coarse location
E3	E2 + target-encoded geo_level_2_id and geo_level_3_id	Value of fine location
E4	E3 + engineered features (floors_x_log_age)	Value of feature engineering
E5	E4 with class_weight='balanced'	Does reweighting help the rare class?
E6	E4 as ordinal (two cumulative binary LRs)	Does exploiting the order help?

The tree ensembles were not put through E0–E6 again. To keep the three-model comparison fair, both use exactly the information of the tuned logistic configuration (E4): structure, geo_level_1_id, target-encoded geo levels 2 and 3, the age flag and the kept engineered feature(s).

4.4 Evaluation metrics

Micro-averaged F1 is the competition metric. For single-label multiclass prediction it equals accuracy. Macro-averaged F1 gives each grade equal weight and so shows performance on the rare grade 1. On the holdout we also report per-class precision and recall and a row-normalised confusion matrix. For the prioritisation analysis, the capture rate at x% is the share of all grade-3 buildings found among the x% of buildings with the highest predicted P(grade 3). Its ratio to x is the lift.

5. Experimental Study and Analysis

5.1 Model 1: logistic-regression experiments

Table 5 summarises the experiments; Appendix A, Figure A1 shows them as a chart.

Table 5. Stratified 5-fold CV results of the logistic-regression experiments on the 80% training split. * = tuned logistic model.

ID	Description	Micro-F1 (mean ± sd)	Macro-F1 (mean ± sd)	Grade-1 recall
E0	Majority-class baseline	0.5689 ± 0.0000	0.2417 ± 0.0000	0.000
E1	Logistic regression, structural features only	0.5908 ± 0.0013	0.4640 ± 0.0021	0.317
E2	E1 + one-hot geo_level_1_id	0.6718 ± 0.0019	0.5917 ± 0.0029	0.357
E3	E2 + target-encoded geo_level_2/3_id	0.7350 ± 0.0012	0.6726 ± 0.0017	0.468
E4	E3 + engineered features (floors_x_log_age)	0.7352 ± 0.0010	0.6732 ± 0.0014	0.470
E5	E4 with class_weight='balanced'	0.6909 ± 0.0011	0.6644 ± 0.0008	0.825
E6	E4 as ordinal logistic (P(y>1), P(y>2))	0.7346 ± 0.0012	0.6728 ± 0.0017	0.470
E4*	**E3 + engineered features (floors_x_log_age), tuned C = 0.3**	**0.7354 ± 0.0010**	**0.6733**	**–**

Structure alone (E1) improves on the 0.569 baseline to 0.591. That is a real gain, but most buildings still end up in the majority class. Coarse location (E2) adds +8.1 pp, and target-encoded fine location (E3) adds a further +6.3 pp, the largest single step in the study. The macro-F1 rises from 0.464 to 0.673 along the same path. Engineered features (E4) change micro-F1 by only +0.02 pp. The raw features already contain most of what a linear model can extract, and the remaining signal is in location rather than geometry. Fold standard deviations are at most 0.0019, so differences of more than about half a percentage point are robust.

Class weighting (E5) lowers micro-F1 by 4.43 pp and changes macro-F1 by -0.88 pp. Grade-1 recall rises from 0.470 to 0.825, and grade-2 recall falls from 0.845 to 0.647. The ordinal model (E6) scores 0.7346 micro-F1 against 0.7352 for the multinomial model. Section 6.1 discusses both. Since micro-F1 is the competition metric and the tuning criterion, the multinomial model with engineered features (E4) was carried forward to tuning.

5.2 Model 1: tuning C

GridSearchCV selected C = 0.3 with CV micro-F1 0.7354 ± 0.0010. The curve is essentially flat (Appendix B, Table B1 and Figure B1): across four orders of magnitude of C, mean micro-F1 varies by only 0.05 percentage points, less than one fold standard deviation. With 208,480 training rows and only 104 model features, the L2 penalty barely matters, so the choice of C is not critical. Strong regularisation (C = 0.01) mainly costs macro-F1 (0.6709 vs 0.6733), because it shrinks the coefficients that separate the rare grade 1.

5.3 Model 1: holdout evaluation

On the untouched holdout (52,121 buildings), the tuned model achieves **micro-F1 = 0.7405** and **macro-F1 = 0.6798**. These are within 0.50 percentage points of the CV estimate, which confirms that model selection did not overfit. Per-class results are in Table 6 and the confusion matrix in Appendix A, Figure A2.

Table 6. Per-class holdout metrics of the tuned logistic model.

Grade	Precision	Recall	F1	Support
1	0.672	0.478	0.559	5,025
2	0.740	0.848	0.790	29,652
3	0.759	0.633	0.691	17,444

Grade 2 is recalled best (84.8%), then grade 3 (63.3%). Grade 1 is hardest (47.8%), and 50.5% of grade-1 buildings are predicted as grade 2. Almost all errors are between adjacent grades. Only 0.48% of grade-3 buildings are predicted as grade 1, and only 1.67% of grade-1 buildings as grade 3. The model therefore rarely makes the costly mistake of calling a destroyed building safe.

5.4 Model 1: prioritisation analysis

To connect the model to the problem statement, we ranked holdout buildings by predicted P(grade 3). We then measured the share of true grade-3 buildings found as inspectors work down the list (Figure 4, Table 7). The figure already includes the two tree ensembles, which are compared in Section 5.7.

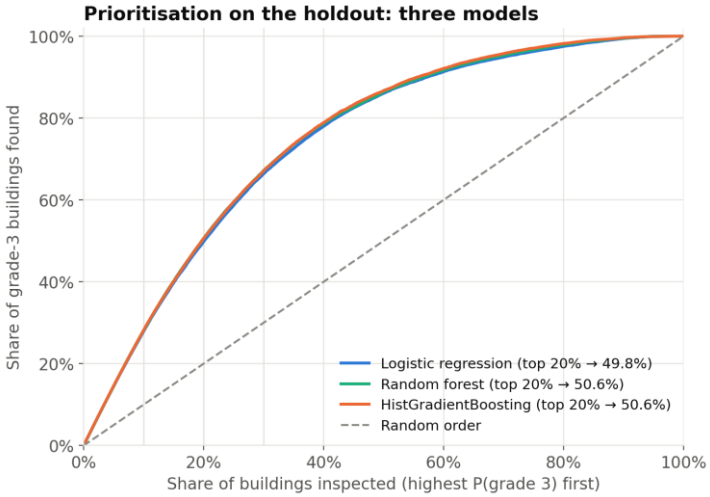


Figure 4. Cumulative share of grade-3 buildings found against the share of buildings inspected, holdout, for the three tuned models. The dashed line is a random inspection order.

Table 7. Capture rates of the tuned logistic model on the holdout (17,444 grade-3 buildings, base rate 33.5%).

Top x% inspected	Buildings inspected	Grade-3 captured	Lift vs random	Grade-3 share within list
10%	5,212	27.9%	2.79×	93.3%
20%	10,424	49.8%	2.49×	83.3%
30%	15,636	66.4%	2.21×	74.1%

Inspecting the top 10% of buildings finds 27.9% of all grade-3 buildings, 2.8 times what a random order finds. Within that top decile, 93.3% of buildings are actually grade 3, against a base rate of 33.5%. At 30% the capture rate is 66.4%. The ROC-AUC for grade 3 against the other grades is 0.866. For an inspection agency, this means the first teams in the field would spend most of their time at buildings that really are destroyed.

5.5 Model 1: interpretation of coefficients

Figure 5 shows the 15 features that most raise and most lower the odds of grade-3 damage in the final model (multinomial grade-3 row). Each is shown as an odds ratio exp(β): per one standard deviation for scaled numeric features, and for presence versus absence for flags and dummy variables. In a softmax model, a dummy's coefficient is relative to the L2-centred average level, so the bars are best read as directions and relative sizes, not exact causal effects. Two groups are excluded from the figure. Location features are discussed below. The other group is 21 flags and dummies present in fewer than 1% of buildings, whose coefficients rest on very few buildings and are unstable. For example, secondary use: school has an odds ratio of 2.08 but applies to only 0.04% of buildings.

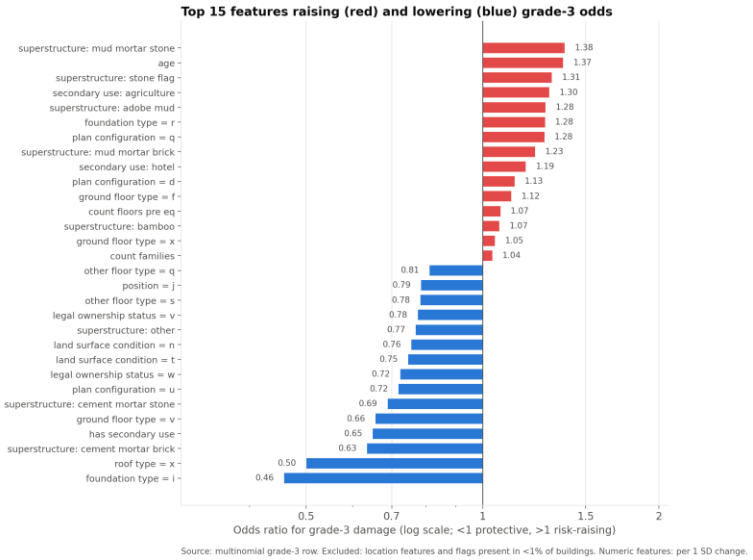


Figure 5. Logistic regression: top 15 risk-raising (red) and risk-lowering (blue) features for grade-3 damage, as odds ratios on a log scale. Location features and flags present in fewer than 1% of buildings are excluded.

The strongest risk-raising features are superstructure: mud mortar stone (OR 1.38), age (OR 1.37), superstructure: stone flag (OR 1.31), secondary use: agriculture (OR 1.30). The strongest protective features are foundation type = i (OR 0.46), roof type = x (OR 0.50), superstructure: cement mortar brick (OR 0.63), has secondary use (OR 0.65). For superstructure materials, holding everything else fixed, the odds ratios are: mud-mortar stone 1.38, stone flag 1.31, adobe mud 1.28, timber 0.88, cement-mortar brick 0.63 and engineered RC 0.84. In plain language, buildings with heavy, brittle walls of stone or mud bonded only by mud mortar break apart under strong shaking. Engineered reinforced-concrete frames and cement-bonded masonry hold together and are much more likely to escape with light damage. Older buildings are also at higher risk. The anonymised foundation and roof codes that the model treats as protective (foundation i, roof x) are the same ones that the exploration (Appendix C, Figure C1) linked to low damage. That the model and the raw data agree, and that both match engineering experience, gives some confidence that the model has learned real effects rather than artefacts. The engineered-RC odds ratio is smaller in magnitude than its raw grade-3 rate would suggest, because foundation, roof and floor types that come with RC construction already carry part of that effect.

Location features, which are excluded from the figure, have the largest coefficients of all (for example geo_level_3_id_3 with OR 22.77, geo_level_3_id_2 with OR 3.11). They describe where the ground shook hardest, not what makes a building vulnerable, so they matter for prediction but give an engineer nothing to act on.

5.6 Models 2 and 3: tuning the tree ensembles

The random forest was tuned over min_samples_leaf ∈ {2, 5}; max_features ∈ {sqrt, 0.25} (4 candidates × 5 folds, 13 minutes). The best setting was max_features = sqrt, min_samples_leaf = 2, with CV micro-F1 0.7440 ± 0.0014 and macro-F1 0.6872. HistGradientBoosting was tuned over learning_rate ∈ {0.05, 0.1, 0.2}; max_leaf_nodes ∈ {15, 31} (6 candidates, 16 minutes). The best setting was learning_rate = 0.1, max_leaf_nodes = 31, with CV micro-F1 0.7452 ± 0.0011 and macro-F1 0.6936. As for C in the logistic model, the grids are fairly flat: the best and worst candidates differ by 0.14 percentage points for the forest and 0.45 for boosting (Appendix B, Tables B2 and B3). The grids are deliberately small so that the whole study runs on an 8 GB laptop, so a wider search could add a little more, but it is unlikely to change the ranking of the models.

5.7 Three-model comparison and choice of the final model

Table 8 and Figure 6 compare the three tuned models. All three use the same features, were tuned on the same folds and were scored once on the same holdout.

Table 8. Three tuned models. CV: stratified 5-fold on the 80% split; holdout: 20%, scored once per model. * = final model (highest CV micro-F1).

Model	CV micro-F1 (mean ± sd)	CV macro-F1	Holdout micro-F1	Holdout macro-F1	Holdout recall g1	Holdout recall g3	Top-20% capture	Grade-3 AUC
Logistic regression	0.7354 ± 0.0010	0.6733	0.7405	0.6798	0.478	0.633	49.8%	0.866
Random forest	0.7440 ± 0.0014	0.6872	0.7502	0.6956	0.509	0.641	50.6%	0.872
HistGradientBoosting *	**0.7452 ± 0.0011**	**0.6936**	**0.7497**	**0.7009**	**0.548**	**0.644**	**50.6%**	**0.875**

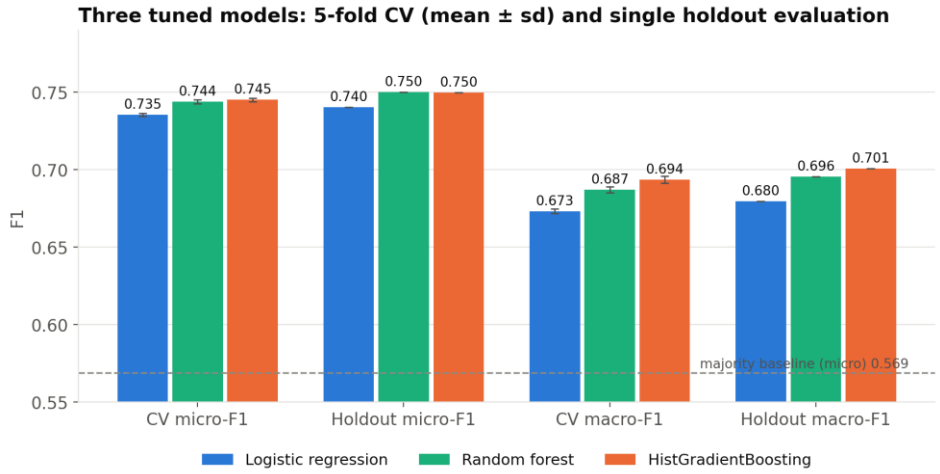


Figure 6. CV (mean ± sd over folds) and holdout micro- and macro-F1 of the three tuned models. The dashed line is the majority-class baseline.

HistGradientBoosting has the highest CV micro-F1 and is therefore our final model. It beats the runner-up (Random forest) by +0.12 pp in CV, winning 5 of 5 folds, and logistic regression by +0.98 pp. The holdout confirms the ranking against logistic regression: 0.7497 against 0.7405 micro-F1 (+0.93 pp) and 0.7009 against 0.6798 macro-F1. The fold standard deviations are about 0.11 percentage points, so a gain of this size is consistent across folds, but it is modest. The trees improve because they can combine features without being told to, for example a material that is only dangerous in strongly shaken areas.

The two ensembles are practically tied. On the holdout the runner-up (Random forest) is even slightly ahead, 0.7502 against 0.7497 micro-F1 (+0.05 pp), a difference well inside the fold-to-fold noise. We keep HistGradientBoosting as the final model because the choice was fixed in advance to rest on cross-validation, which averages five validation sets instead of one; switching after seeing the holdout would turn the holdout into a selection set. HistGradientBoosting also has the higher macro-F1 on the holdout (0.7009 against 0.6956).

For the inspection use case the three rankings are close (Figure 4): the top 20% of the list captures 49.8% (Logistic regression), 50.6% (Random forest), 50.6% (HistGradientBoosting) of grade-3 buildings, and the grade-3 ROC-AUC ranges from 0.866 to 0.875. Any of the three models would therefore send the first inspection teams to largely the same high-risk buildings.

Figure 7 shows the holdout confusion matrix of HistGradientBoosting. As for logistic regression, almost all errors are between adjacent grades: 43.8% of grade-1 buildings are predicted as grade 2 and 35.0% of grade-3

buildings as grade 2, while only 0.60% of grade-3 buildings are predicted as grade 1. Grade-1 recall is 54.8% (logistic regression: 47.8%) and grade-3 recall is 64.4% (63.3%).

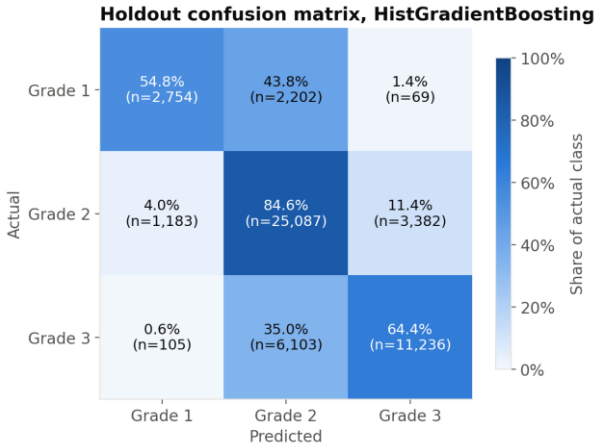


Figure 7. Row-normalised holdout confusion matrix of the final model, HistGradientBoosting (counts in brackets).

Tree ensembles have no coefficients, so we complement the odds ratios with *permutation importance* (Breiman, 2001): the drop in micro-F1 when one original input column is randomly shuffled, measured on a random subsample of 10,000 holdout buildings with five repeats. Because the whole pipeline is permuted, geo_level_3_id, for example, covers all three of its target-encoded columns. We prefer this to impurity-based importance, which is biased toward continuous and high-cardinality columns. Figure 8 shows the results for all three models. For HistGradientBoosting the most important columns are geo level 3 id (0.111), geo level 2 id (0.044), geo level 1 id (0.017), age (0.010). All three models rank the fine location levels (geo levels 3 and 2) first, followed by age and construction attributes such as superstructure material, which agrees with the exploration and with the logistic coefficients. Permutation importance tells us how much a model relies on a column, not in which direction it pushes the risk; that direction is exactly what the logistic odds ratios provide.

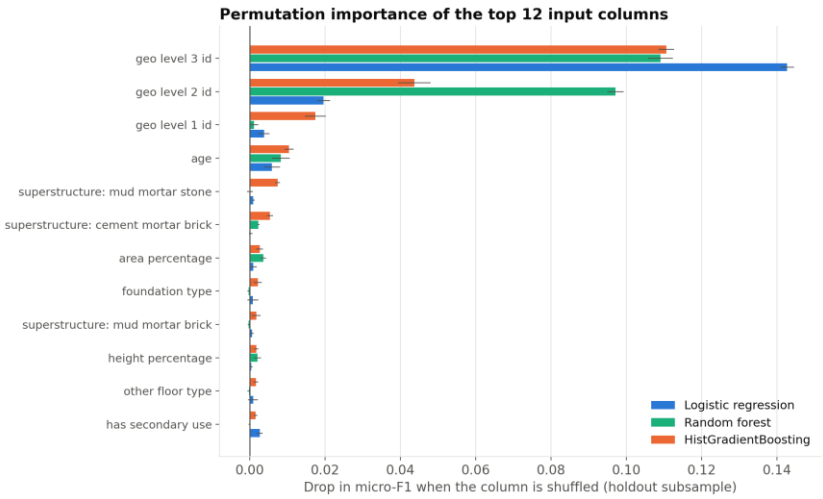


Figure 8. Permutation importance (drop in holdout micro-F1 when a column is shuffled; mean ± sd over 5 repeats on 10,000 holdout buildings) of the 12 most important input columns.

6. Highlights

6.1 Ordinal versus multinomial logistic regression

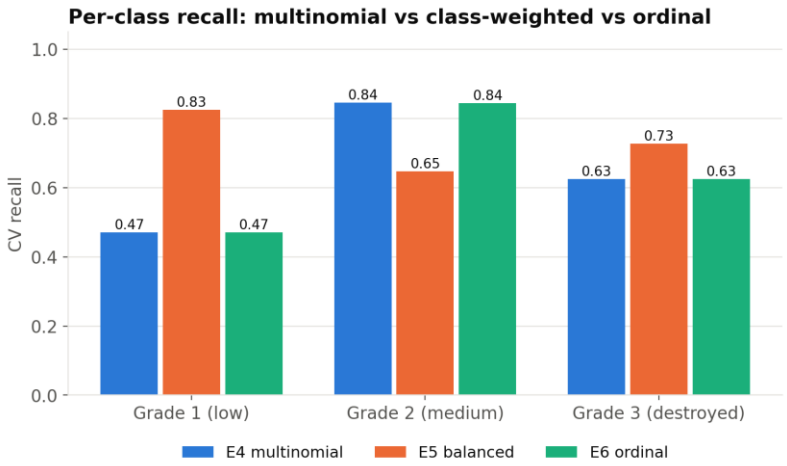


Figure 9. Per-class CV recall for the multinomial (E4), class-balanced (E5) and ordinal (E6) models.

Exploiting the ordering of damage grades seems natural, but on these data it **did not help**: the ordinal model changed micro-F1 by -0.06 pp and macro-F1 by -0.04 pp compared with the multinomial model on identical features (Figure 9). We see three reasons for this negative result. (i) With only three classes, the multinomial model has just one more set of coefficients than the ordinal model and can learn an ordered structure by itself. Its fitted probabilities already put very little mass on the non-adjacent grade (Section 5.3). (ii) Our ordinal model fits the two thresholds independently, so it gains nothing from sharing slopes. It can also produce crossing cumulative probabilities, which we have to clip. (iii) Ordinal structure mostly helps by avoiding large errors (1 ↔ 3), but micro-F1 counts every error the same, and large errors were already rare. This is still a useful finding: for this problem the simpler multinomial model is sufficient.

Class weighting (E5) did what it is designed to do for recall. Reweighting the rare grade 1 raised its recall from 0.47 to 0.83 and grade-3 recall from 0.63 to 0.73. The cost was grade-2 recall, which fell from 0.84 to 0.65. Micro-F1 therefore fell (-4.43 pp), and macro-F1 also fell (-0.88 pp). Macro-F1 falling is less obvious than micro-F1 falling. The extra grade-1 predictions are taken mostly from the large grade-2 class, so grade-1 precision drops faster than its recall rises, and grade-2 F1 falls as well. Whether that trade is worth making is a policy decision rather than a modelling one. An agency that cares about quickly clearing low-damage buildings, or about flagging more grade-3 buildings when acting on predicted labels rather than on a ranking, might prefer it. For the competition metric, the unweighted model is better.

6.2 Cross-fitted target encoding of location

The finest location level has 11,595 values, and level 2 has 1,414. One-hot encoding them would add 13,009 sparse columns, most with only a handful of buildings, and an L2-penalised linear model would either overfit them or shrink them to nothing. Multiclass target encoding compresses each level into three smoothed damage rates, which directly answer the question 'how badly was this neighbourhood hit?'. It produced the largest gain in the study, +6.3 pp micro-F1 and +8.1 pp macro-F1 over E2. Two safeguards keep this honest. First, cross-fitting inside fit_transform means that no training building's encoding uses its own label. Second, the encoder is part of the pipeline and so is refitted inside every outer CV fold and inside GridSearchCV. The close agreement between CV and holdout scores (0.7354 vs 0.7405) is consistent with there being no leakage.

The same encoder feeds the two tree ensembles, and for all three models the location columns are among the most important inputs (Figure 8).

6.3 Accuracy versus interpretability

Adding two tree ensembles answers a question the logistic-only study could not: how much accuracy does transparency cost? On identical features, the best tree ensemble (HistGradientBoosting) gains +0.98 pp CV micro-F1 and +0.93 pp holdout micro-F1 over logistic regression, while the prioritisation curves are almost identical (50.6% against 49.8% of grade-3 buildings in the top 20%). The price of transparency is therefore real but small. We use the most accurate model for the competition submission, but for explaining risk to engineers and to the public the logistic model's odds ratios remain the better tool, and the permutation importances show that both kinds of model rely on the same signals.

7. Limitations

- **Linear model versus black boxes.** Logistic regression cannot represent interactions it is not given, such as material × region. The tree ensembles can, but they offer only importance scores, not signed, per-unit effects. The original brief excluded tree models; they were added later at the course's request, with the same features and folds, but the tree study is less deep than the logistic one (no separate feature experiments, small tuning grids).
- **Location as a proxy.** Region ids stand in for shaking intensity, soil and distance to the fault. The model would not transfer to a different earthquake, and 0.38% of test buildings fall in unseen local areas and receive only the global prior.
- **Labels and features recorded after the event.** Damage grades come from survey teams and may be noisy near grade boundaries. Some attributes, such as floor count before the earthquake, were reconstructed after it.
- **Anonymised categories.** Letter codes limit how far the coefficients can be explained in engineering terms.
- **Associational, not causal.** Odds ratios describe associations conditional on the other features. Collinear inputs (height and floors) share credit, and the softmax parameterisation makes dummy coefficients relative to an average level.
- **Single split.** The holdout estimate comes from one 20% split. CV standard deviations are small, but no confidence interval for the holdout score was computed, and the differences between the three models are only a few fold standard deviations.

8. Future Directions

- **Physical hazard features.** Join measured peak ground acceleration (USGS ShakeMap) and distance to the rupture, replacing region ids with physics that transfers to future events.
- **Interactions and splines.** Add material × region, floors × age and spline terms for age and area, which keeps the model linear in its parameters and interpretable.
- **Proper cumulative-link model.** Fit a proportional-odds model with shared slopes (for example statsmodels OrderedModel) and test the proportional-odds assumption.
- **Cost-sensitive thresholds and calibration.** Choose decision thresholds from inspection capacity and the relative cost of missing a destroyed building, and calibrate probabilities (Platt or isotonic).
- **Uncertainty and validation.** Bootstrap confidence intervals for holdout metrics, and spatially grouped CV (by geo_level_2) to estimate performance in unseen areas.

- **Explaining and extending the ensembles.** Use SHAP values to give the tree models per-building explanations, search larger grids (or try LightGBM/CatBoost), and package the final pipeline as a triage tool for field teams.

9. Conclusion

We asked whether a building's structure and location can predict its earthquake damage well enough to prioritise inspections. With a leakage-safe pipeline and three classifiers, the answer is a qualified yes. The tuned logistic regression reaches a holdout micro-F1 of 0.740 (macro-F1 0.680) against a 0.569 baseline; the random forest reaches 0.750 and HistGradientBoosting 0.750. Chosen by cross-validation, HistGradientBoosting is our final model, +0.98 pp ahead of logistic regression in CV and +0.93 pp on the holdout. Its ranking finds 50.6% of destroyed buildings in the first 20% of inspections, and all three models rank buildings almost equally well. Location, encoded with cross-fitted target encoding, contributed the most for every model. Construction material contributed the most interpretable signal: mud-mortar stone raises risk and engineered RC lowers it. An ordinal reformulation did not improve on the multinomial model, and class weighting traded overall accuracy for recall of the rare class. The small gap between the linear model and the tree ensembles suggests that the largest further gains will come from better information, such as physical shaking data, rather than from more complex models.

References

Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5–32.

DrivenData. (2019). Richter's Predictor: Modeling Earthquake Damage [Data set and competition]. https://www.drivendata.org/competitions/57/nepal-earthquake/

Frank, E., & Hall, M. (2001). A simple approach to ordinal classification. In Proceedings of the 12th European Conference on Machine Learning (ECML), LNCS 2167, 145–156. Springer.

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. Annals of Statistics, 29(5), 1189–1232.

Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning (2nd ed.). Springer.

Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). Applied Logistic Regression (3rd ed.). Wiley.

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. Computing in Science & Engineering, 9(3), 90–95.

McCullagh, P. (1980). Regression models for ordinal data. Journal of the Royal Statistical Society: Series B, 42(2), 109–142.

McKinney, W. (2010). Data structures for statistical computing in Python. In Proceedings of the 9th Python in Science Conference, 56–61.

Micci-Barreca, D. (2001). A preprocessing scheme for high-cardinality categorical attributes in classification and prediction problems. ACM SIGKDD Explorations, 3(1), 27–32.

National Planning Commission. (2015). Nepal Earthquake 2015: Post Disaster Needs Assessment, Vol. A: Key Findings. Government of Nepal, Kathmandu.

Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

Appendix

A. Full cross-validation results (logistic regression)

Table 9. All experiments: 5-fold CV mean ± sd; recall is the per-class CV mean; max solver iterations over the 5 folds.

ID	Micro-F1	Macro-F1	Recall g1	Recall g2	Recall g3	Max solver iter
E0	0.5689 ± 0.0000	0.2417 ± 0.0000	0.000	1.000	0.000	0
E1	0.5908 ± 0.0013	0.4640 ± 0.0021	0.317	0.880	0.177	143
E2	0.6718 ± 0.0019	0.5917 ± 0.0029	0.357	0.816	0.517	162
E3	0.7350 ± 0.0012	0.6726 ± 0.0017	0.468	0.845	0.625	258
E4	0.7352 ± 0.0010	0.6732 ± 0.0014	0.470	0.845	0.625	292
E5	0.6909 ± 0.0011	0.6644 ± 0.0008	0.825	0.647	0.727	259
E6	0.7346 ± 0.0012	0.6728 ± 0.0017	0.470	0.844	0.625	175

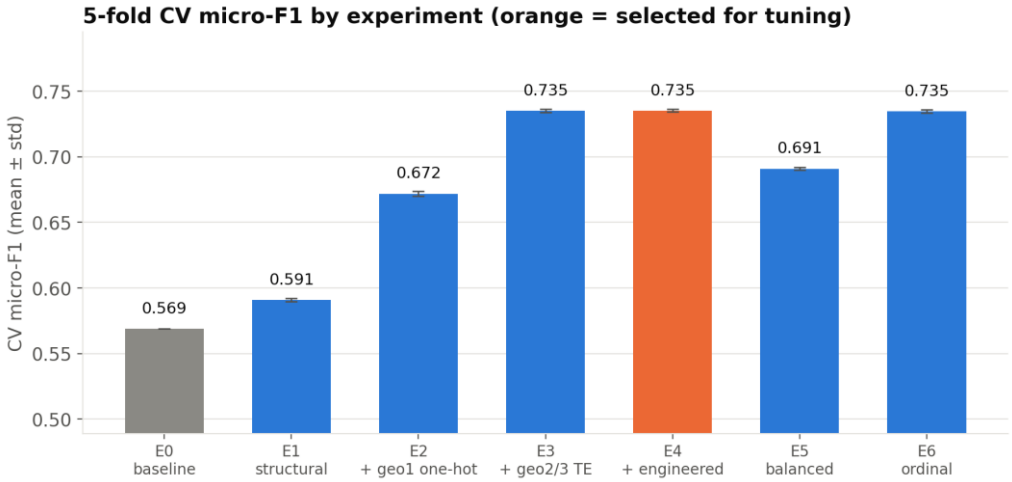


Figure A1. CV micro-F1 by logistic-regression experiment (error bars: ±1 sd over folds). Orange marks the configuration selected for tuning.

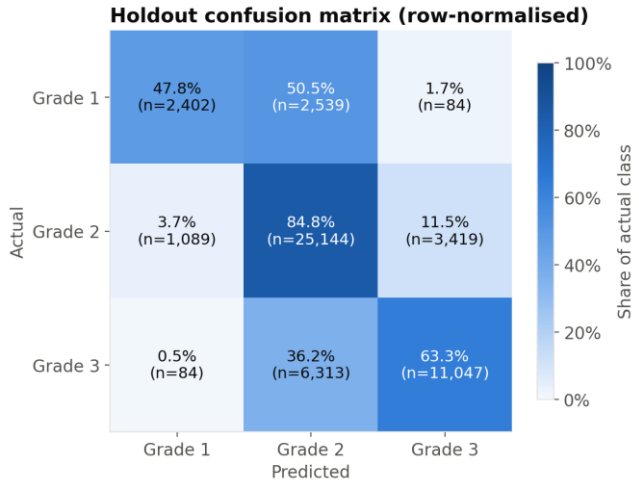


Figure A2. Row-normalised holdout confusion matrix of the tuned logistic model (counts in brackets).

B. Tuning grids

Table B1. Model 1: GridSearchCV results for logistic regression (E4) over C.

C	Micro-F1 (mean)	Micro-F1 (sd)	Macro-F1 (mean)	Max solver iterations
0.01	0.7349	0.0011	0.6709	96
0.03	0.7352	0.0012	0.6723	110
0.1	0.7353	0.0011	0.6730	175
0.3	0.7354	0.0010	0.6733	232
1	0.7352	0.0010	0.6732	292
3	0.7351	0.0009	0.6731	317
10	0.7351	0.0010	0.6732	324
30	0.7350	0.0010	0.6731	336
100	0.7351	0.0011	0.6732	337

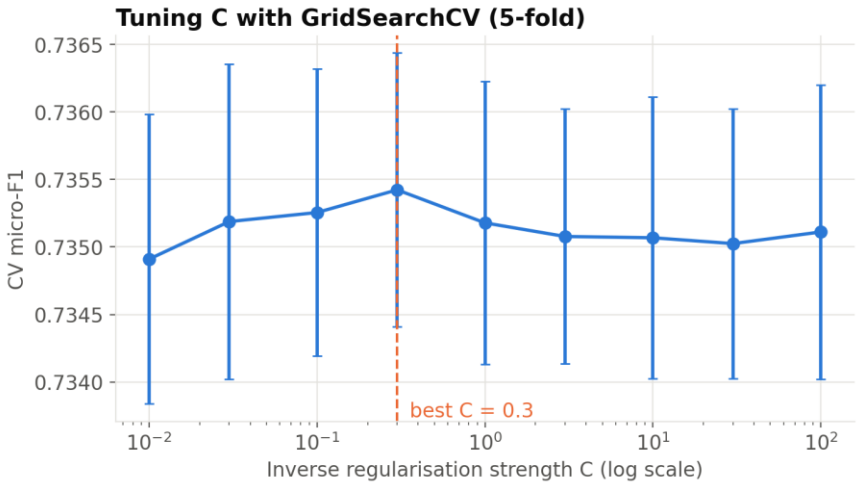


Figure B1. CV micro-F1 as a function of C (log scale). The dashed line marks the selected value.

Table B2. Model 2: GridSearchCV results for Random forest (5 folds, scored by f1_micro). Fixed settings: n_estimators = 150, max_samples = 0.5, bootstrap = True, criterion = gini. Best: max_features = sqrt, min_samples_leaf = 2.

min_samples_leaf	max_features	Micro-F1 (mean)	Micro-F1 (sd)	Macro-F1 (mean)	Fit time per fold (s)
2	sqrt	0.7440	0.0014	0.6872	25
5	sqrt	0.7430	0.0014	0.6841	32
2	0.25	0.7430	0.0017	0.6869	45
5	0.25	0.7425	0.0011	0.6851	38

Table B3. Model 3: GridSearchCV results for HistGradientBoosting (5 folds, scored by f1_micro). Fixed settings: loss = log_loss, max_iter = 300, early_stopping = False, l2_regularization = 0.0. Best: learning_rate = 0.1, max_leaf_nodes = 31.

learning_rate	max_leaf_nodes	Micro-F1 (mean)	Micro-F1 (sd)	Macro-F1 (mean)	Fit time per fold (s)

0.05	15	0.7432	0.0017	0.6903	31
0.05	31	0.7444	0.0010	0.6922	32
0.1	15	0.7441	0.0009	0.6922	26
0.1	31	0.7452	0.0011	0.6936	30
0.2	15	0.7426	0.0012	0.6897	23
0.2	31	0.7406	0.0013	0.6885	32

C. Additional exploration figures

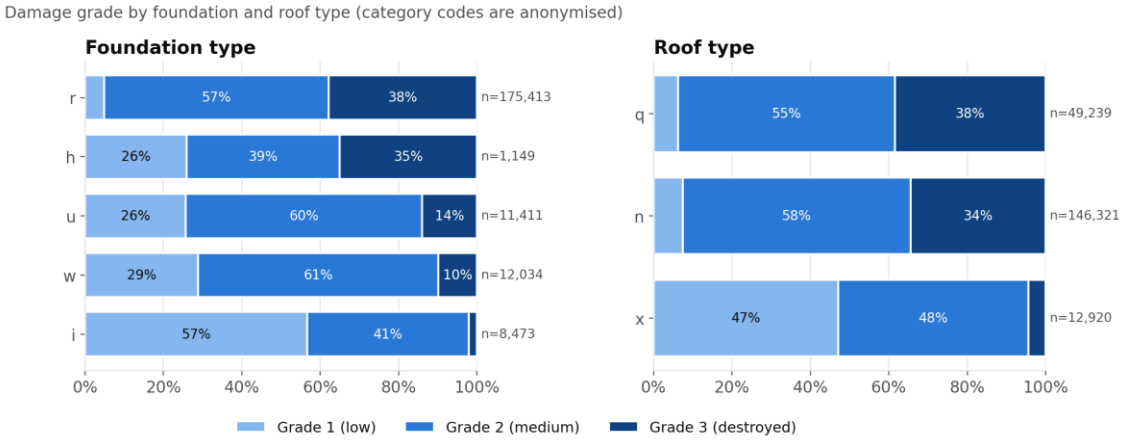


Figure C1. Damage-grade mix by foundation type (left) and roof type (right), training split. Bars are sorted by grade-3 share; n is the number of buildings.

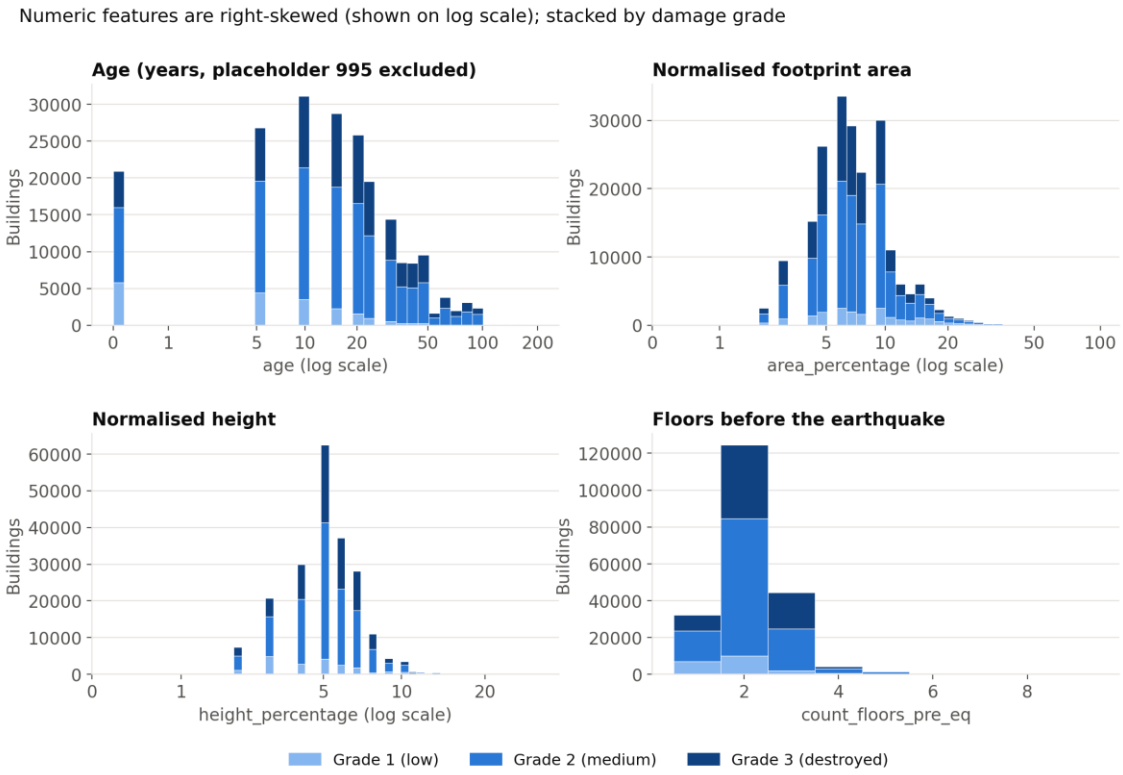


Figure C2. Distributions of age, area, height (log scale) and floors, stacked by damage grade. Placeholder ages are excluded.

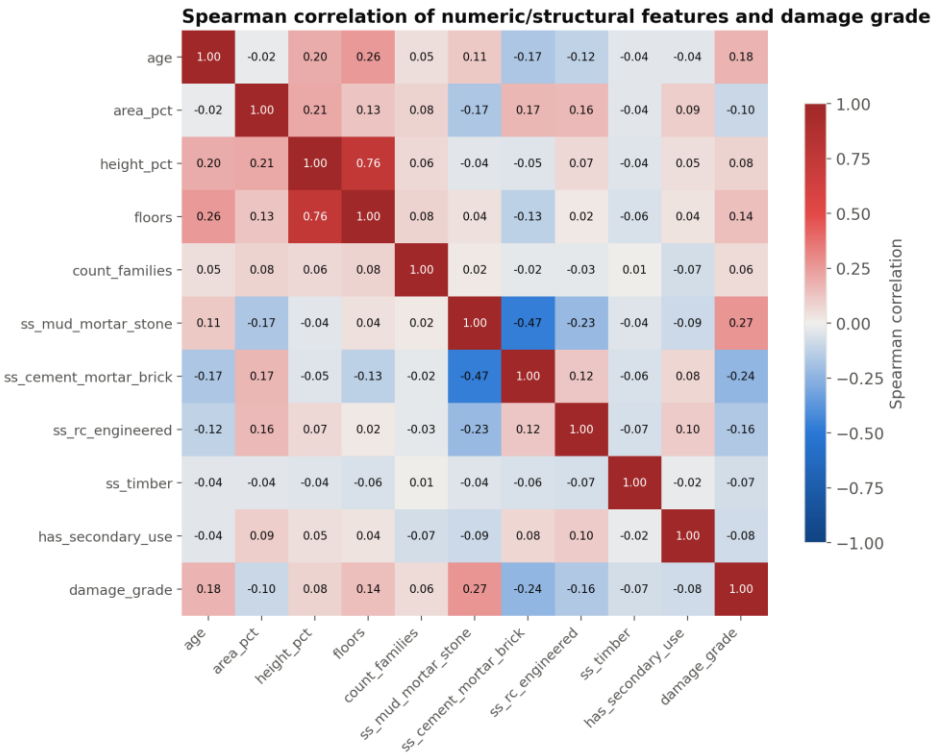


Figure C3. Spearman correlations among numeric and selected material features and the damage grade.

D. Reproducibility

Environment: Python 3.11.14, scikit-learn 1.5.2 (Pedregosa et al., 2011), pandas 2.2.3 (McKinney, 2010), NumPy 2.1.3, matplotlib 3.9.2 (Hunter, 2007), seaborn 0.13.2. random_state = 42 everywhere; StratifiedKFold(n_splits=5, shuffle=True, random_state=42). The full pipeline runs with **python -m src.run_all** (about 59.4 minutes for the notebook). Every number in this report is read from outputs/results.json, which the notebook writes. The submission file outputs/submission.csv comes from the final model (HistGradientBoosting, refitted on all 260,601 labelled buildings) and has 86,868 rows with columns building_id, damage_grade (integers 1–3). Its predicted class shares are grade 1 7.8%, grade 2 64.2% and grade 3 28.1%.

E. Contribution statement

[To be completed by the group: one line per member describing their contribution.]